

< Previous

 ✓

 ✓

 ✓



Next >

Notifications 

4. Eigenvalues of symmetric matrices using MATLAB

 Bookmark this page

Upgrade your course today

✓

You are taking the free version of this course.

✓

Upgrade today and, upon passing, receive your digital certificate signed by MIT faculty to highlight the knowledge and skills you’ve gained from this MITx course.

Upgrade

Eigenvectors of a symmetric matrix (External resource)

(1.0 points possible)

the length of the projection of one vector onto the other. If this projection is zero, this says that the vectors are perpendicular in the plane that they span.

Note that we can write the dot product of two column vectors of length n by first taking the transpose of the first vector $\mathbf{v}^T \mathbf{w}$ and then the product is the regular matrix product. This is well defined since $\mathbf{v}^T$ is a row vector that is $1 \times n$ and $\mathbf{w}$ is a column vector that is $n \times 1$, and thus the product is a real number, a 1×1 matrix. Thus the length squared of a vector with real entries is $\mathbf{v}^T \mathbf{v}$. What happens if the entries of a vector are complex (as they can be for an eigenvector)? In this case, length squared of a vector is defined by taking the complex conjugate transpose of the first vector $\bar{\mathbf{v}}^T \mathbf{v}$.

Note that in matlab, the prime notation which takes the transpose of a matrix is automatically taking conjugate transpose! So to take the dot product of any two vectors in matlab (with real or complex entries), we type

```
v'*w
```

There is another useful result in linear algebra which says that the eigenvectors of any distinct eigenvalues of any symmetric matrix are orthogonal. MATLAB gives us a simple way to test whether or not this is a reasonable claim. MATLAB can be a useful tool for testing out hypotheses, however, you must remember that numerical experiments are never a substitute for a full proof.

In order to understand the experiment below, it is useful to note that when you use the eig() command in MATLAB, the eigenvectors it gives you are of unit length. Therefore all of the dot products given will have magnitude less than 1.

Script ?

 Save  Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 % Firstly generate a random 100x100 matrix A of numbers randomly chosen between -10 and 10.
2 % Note that rand() generates random numbers between 0 and 1, so we've modified the range
3 % by first subtracting 0.5 from each entry to get randomly generated entries between -0.5 and 0.5
4 % then multiplying by 20 so that the numbers lie between -10 and 10.
5
6 A = (rand(100)-0.5)*20;
7 % Now calculate the symmetric part of the matrix A. Call this matrix B
8 B = 0.5*(A + A');
9 % Now calculate the eigenvalues and eigenvectors of A using eig(). Store the results in S1 and D1
10 [S1, D1] = eig(A);
11 % Now calculate the eigenvalues and eigenvectors of B using eig(). Store the results in S2 and D2
12 [S2, D2] = eig(B);
13
14 % We will now create a scatter plot of the first eigenvector of A and B dotted with each other
15 % of A and B using the script you have just completed.
16 % If you have filled it in correctly, then you should see that the dot products of the first
17 % eigenvectors of A and B are scattered across the complex plane,
18 % while the dot product of one eigenvector of B with any other eigenvector of B is zero.
19 figure(1)
20 hold on
21 for m=2:100
22     plot(real(S1(:,m)'*S1(:,1)),imag(S1(:,m)'*S1(:,1)), 'b*')
23 end
24 xlabel('Real part')
25 ylabel('Imaginary part')
26 xlim([-0.5, 0.5])
27 ylim([-0.5, 0.5])
28 title('Dot products of eigenvectors of A')
29 figure(2)
30 hold on
31 for m=2:100
32     plot(real(S2(:,m)'*S2(:,1)),imag(S2(:,m)'*S2(:,1)), 'r*')
33 end
34 xlabel('Real part')
35 ylabel('Imaginary part')
36 xlim([-0.5, 0.5])
37 ylim([-0.5, 0.5])
38 title('Dot products of eigenvectors of B')
```

 Run Script 

✔ Symmetric part correct

✔ S1 correct

✔ D1 correct

✔ S2 correct

✔ D2 correct

For your entertainment, here are proofs that the eigenvalues of a symmetric matrix are real, and that any two eigenvectors of distinct eigenvalues are orthogonal. These proofs rely on two facts about how matrix multiplication interacts with complex conjugation and transposes. Let **A** and **B** be two matrices, possibly with complex entries.

- $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$
- $\overline{\mathbf{AB}} = (\overline{\mathbf{A}})(\overline{\mathbf{B}})$
- $(\overline{\mathbf{AB}})^T = \overline{\mathbf{B}}^T \overline{\mathbf{A}}^T$

Proof that eigenvalues of a real valued symmetric matrix are real

Let **M** be a symmetric real valued matrix. Then in particular $\overline{\mathbf{M}} = \mathbf{M}$ and $\mathbf{M}^T = \mathbf{M}$.

Let **v** be a nonzero eigenvector of **M** with eigenvalue λ . Note that the length of any vector is nonnegative, so in particular $|\mathbf{M}\mathbf{v}| \geq 0$. Writing this in terms of matrix products we get

$$\begin{aligned} |\mathbf{M}\mathbf{v}|^2 &= ((\overline{\mathbf{M}\mathbf{v}})^T)(\mathbf{M}\mathbf{v}) \\ &= (\overline{\mathbf{v}}^T \overline{\mathbf{M}}^T)(\mathbf{M}\mathbf{v}) \\ &= \overline{\mathbf{v}}^T \mathbf{M}^2 \mathbf{v} \\ &= \overline{\mathbf{v}}^T \lambda^2 \mathbf{v} = \lambda^2 |\mathbf{v}|^2 \end{aligned}$$

The last line is true since if $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$, then $\mathbf{M}^2\mathbf{v} = \mathbf{M}(\lambda\mathbf{v}) = \lambda^2\mathbf{v}$.

Note that we now know that $|\mathbf{M}\mathbf{v}|^2 = \lambda^2 |\mathbf{v}|^2 \geq 0$. Since $|\mathbf{v}|^2 > 0$, and $|\mathbf{M}\mathbf{v}|^2 \geq 0$, it follows that $\lambda^2 \geq 0$ as well. Therefore λ is real.

[Hide](#)

Proof that eigenvectors corresponding to different eigenvalues of a symmetric matrix are orthogonal

Let **M** be a symmetric real valued matrix. (Then in particular $\overline{\mathbf{M}} = \mathbf{M}$ and $\mathbf{M}^T = \mathbf{M}$.)

Let **v**₁ be an eigenvector of **M** with eigenvalue λ_1 , and **v**₂ be an eigenvector of **M** with eigenvalue λ_2 for distinct eigenvalues λ_1 and λ_2 .

To show that **v**₁ and **v**₂ are orthogonal, we will express $\overline{\mathbf{v}}_2^T \mathbf{M}\mathbf{v}_1$ in two ways.

$$\begin{aligned} \overline{\mathbf{v}}_2^T (\mathbf{M}\mathbf{v}_1) &= \overline{\mathbf{v}}_2^T (\lambda_1 \mathbf{v}_1) \\ &= \lambda_1 \overline{\mathbf{v}}_2^T \mathbf{v}_1 \end{aligned}$$

Using associativity to move the parentheses on the opposite side we can also write this expression as

$$(\overline{\mathbf{v}}_2^T \mathbf{M}) \mathbf{v}_1 = (\overline{\mathbf{v}}_2^T \overline{\mathbf{M}}^T) \mathbf{v}_1$$

© MITx Online. All rights reserved except where noted.

[About Us](#)

[Terms of Service](#)

[Privacy Policy](#)

[Honor Code](#)

[Accessibility](#)

$$\lambda_1 \overline{\mathbf{v}}_2^T \mathbf{v}_1 = \lambda_2 \overline{\mathbf{v}}_2^T \mathbf{v}_1.$$

Since $\lambda_1 \neq \lambda_2$, this equality can only hold if $\overline{\mathbf{v}}_2^T \mathbf{v}_1 = 0$. Therefore $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal.

[Hide](#)

4. Eigenvalues of symmetric matrices using MATLAB

Hide Discussion

Topic: Unit 2: Linear Algebra, Part 2 / 4. Eigenvalues of symmetric matrices using MATLAB